Bitcoin Market Sentiment vs Trader Performance Analysis

1. Introduction

Cryptocurrency markets are highly influenced by investor psychology, emotional sentiment, and speculative behavior. Market sentiment indicators such as the Fear & Greed Index provide valuable insight into trader confidence and overall market conditions.

This project analyzes the relationship between Bitcoin market sentiment and trader performance using:

- Historical Hyperliquid trading data
- Bitcoin Fear & Greed Index dataset

The goal of this analysis is to identify hidden behavioral patterns, evaluate profitability trends, analyze trading activity, and derive insights that can support smarter trading strategies and better risk management.

2. Objective

The primary objectives of this analysis are:

- To evaluate trader profitability across different market sentiment conditions
- To analyze how sentiment impacts trader win rates
- To study changes in trading activity and risk exposure
- To identify patterns among top-performing traders
- To statistically evaluate whether sentiment significantly affects trading performance

3. Dataset Description

3.1 Historical Trader Dataset

The Hyperliquid trading dataset contains detailed trader-level transaction information.

Important Features

Feature	Description
Account	Trader identifier
Coin	Cryptocurrency traded

Feature	Description
Execution Price	Trade execution price
Size USD	Trade size in USD
Side	Buy/Sell direction
Timestamp IST	Trade timestamp
Closed PnL	Realized profit/loss

3.2 Fear & Greed Dataset

The Fear & Greed dataset contains Bitcoin market sentiment values and classifications.

Important Features

Feature	Description
date	Market date
value	Fear & Greed index value
classification	Sentiment category

Sentiment Categories

- Extreme Fear
- Fear
- Neutral
- Greed
- Extreme Greed

4. Data Preprocessing

Several preprocessing steps were performed before analysis:

- Cleaned and standardized dataset columns
- Converted timestamps into datetime format
- Extracted date values for merging datasets
- Removed inconsistencies and null values

- Merged trading data with sentiment data using common dates

The final merged dataset contained approximately 211,224 trading records linked with corresponding market sentiment conditions.

5. Feature Engineering

Additional analytical features were created to improve performance analysis.

5.1 Profitability Indicator

Trades were classified as profitable or non-profitable using Closed PnL values.

5.2 Risk Exposure

Risk exposure was estimated using trade size values in USD.

5.3 Absolute Profit/Loss

Absolute PnL values were calculated to measure trading volatility and magnitude of outcomes.

6. Exploratory Data Analysis

Initial exploratory analysis revealed large variation in trade sizes, trader profitability, and market participation.

Summary Statistics

Metric	Value
Total Records	211,224
Average Trade Size (USD)	5,639.45
Median Trade Size (USD)	597.05
Maximum Trade Size (USD)	3,921,430.72
Average Closed PnL	48.75
Maximum Closed PnL	135,329.09
Minimum Closed PnL	-117,990.10

Key Findings

- The dataset contains both small retail-scale trades and extremely large institutional-scale positions.
- The large difference between median and maximum trade size indicates significant trading imbalance across participants.
- Profit and loss values show high volatility, highlighting the speculative nature of cryptocurrency markets.

index	Execution Price	Size Tokens	Size USD	Timestamp IST	Start Position	Closed PnL	Order ID
count	211224.0	211224.0	211224.0	211224	211224.0	211224.0	211224.0
mean	11414.723349993757	4623.364978782207	5639.4512103738225	2025-01-31 12:04:22.915009792	-29946.2488391739	48.74900079269401	69653876008.97021
min	4.53e-06	8.74e-07	0.0	2023-05-01 01:06:00	-14334629.0	-117990.1041	173271100.0
25%	4.8547	2.94	193.79	2024-12-31 21:00:45	-376.23107500000003	0.0	59838527992.75
50%	18.28	32.0	597.0450000000001	2025-02-24 18:55:00	84.727931935	0.0	74429390066.0
75%	101.58	187.9025	2058.96	2025-04-02 18:22:00	9337.2775	5.79279675	83355430544.0
max	109004.0	15822438.0	3921430.72	2025-05-01 12:13:00	30509482.0	135329.0901	90149230487.0
std	29447.654868348793	104272.889530004	36575.138546399205	NaN	673807.4237357653	919.1648281992385	18357525271.925747

7. Analysis 1 — Trader Profitability vs Market Sentiment

The first analysis examined how trader profitability changes across different market sentiment conditions.

Average Closed PnL by Sentiment

Market Sentiment	Average Closed PnL
Neutral	34.31
Extreme Fear	34.54
Greed	42.74
Fear	54.29
Extreme Greed	67.89

Key Findings

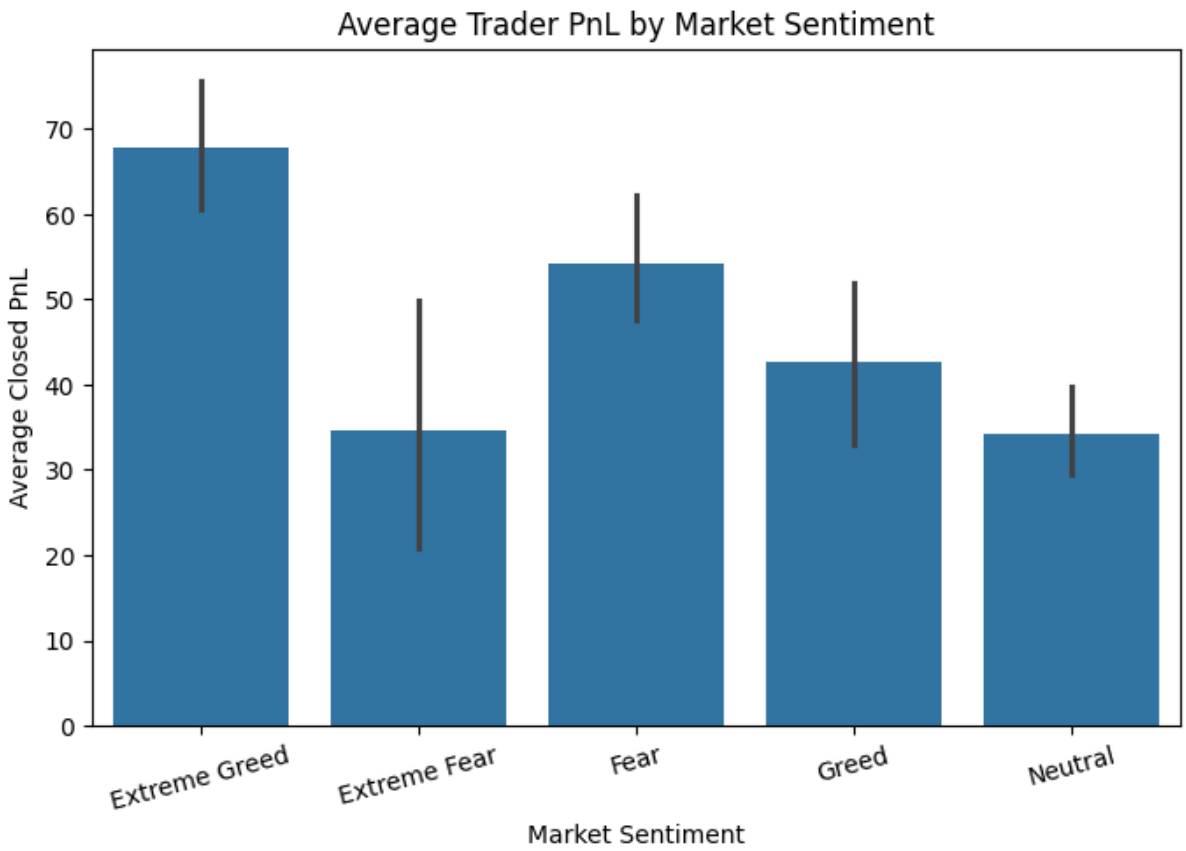
- Extreme Greed produced the highest average profitability at approximately **67.89**.
- Fear conditions also showed relatively strong profitability with an average Closed PnL of **54.29**.
- Neutral and Extreme Fear conditions produced the lowest average profitability values.

Interpretation

The findings suggest that emotionally intense market conditions create stronger trading opportunities and larger price movements. Bullish sentiment environments, particularly

Extreme Greed phases, may enhance short-term trading profitability due to stronger momentum and speculative activity.

However, these periods may also expose traders to increased volatility and risk.



8. Analysis 2 — Trader Win Rate

This analysis evaluated the percentage of profitable trades under different market sentiment conditions.

Win Rate by Sentiment

Market Sentiment	Win Rate (%)
Extreme Fear	37.06%
Greed	38.48%
Neutral	39.70%
Fear	42.08%
Extreme Greed	46.49%

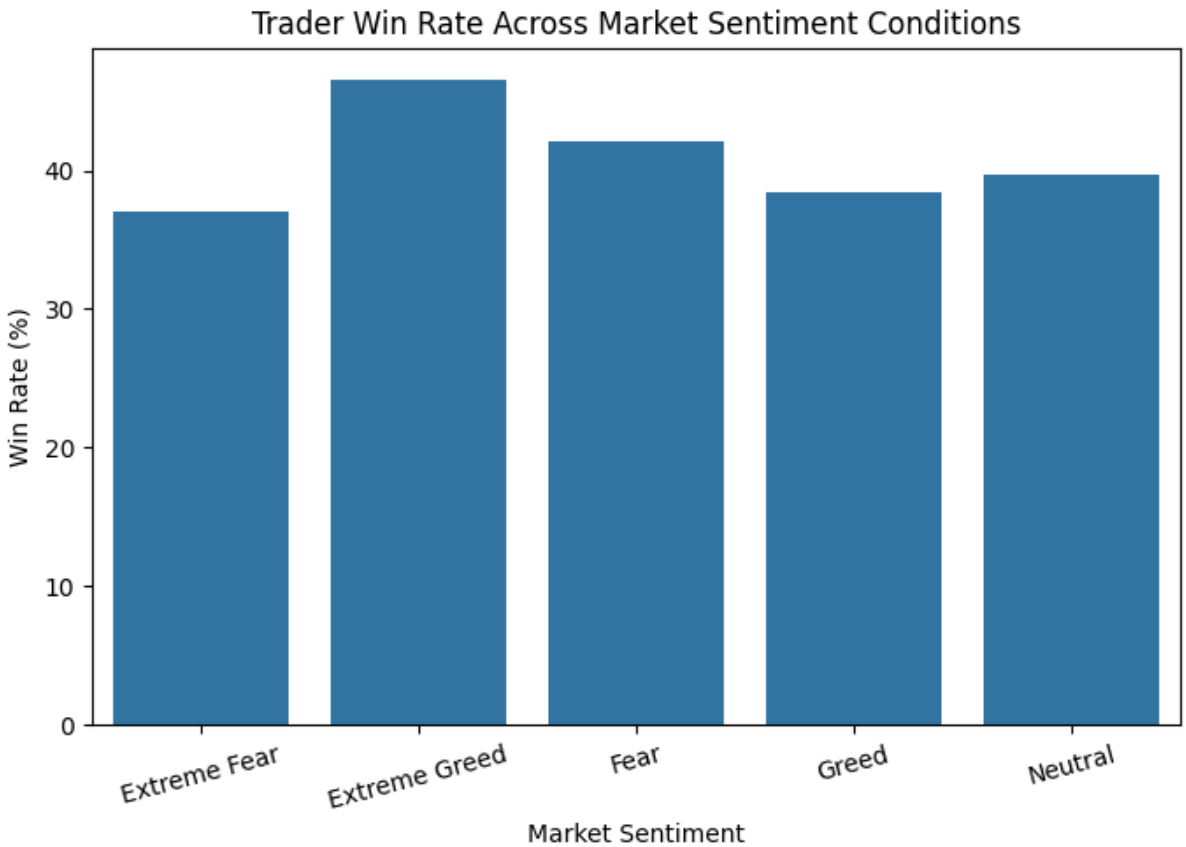
Key Findings

- Extreme Greed markets produced the highest trader win rate at **46.49%**.
- Fear conditions also demonstrated relatively strong performance with win rates above **42%**.
- Extreme Fear conditions generated the lowest trading success rate.

Interpretation

Bullish sentiment conditions appear to improve short-term trading consistency and increase the probability of profitable outcomes. Traders may perform better during optimistic markets due to stronger directional momentum and increased market participation.

In contrast, bearish environments introduce greater uncertainty and inconsistent price action.



9. Analysis 3 — Trading Activity

Trading activity analysis was performed to understand how sentiment affects trader participation.

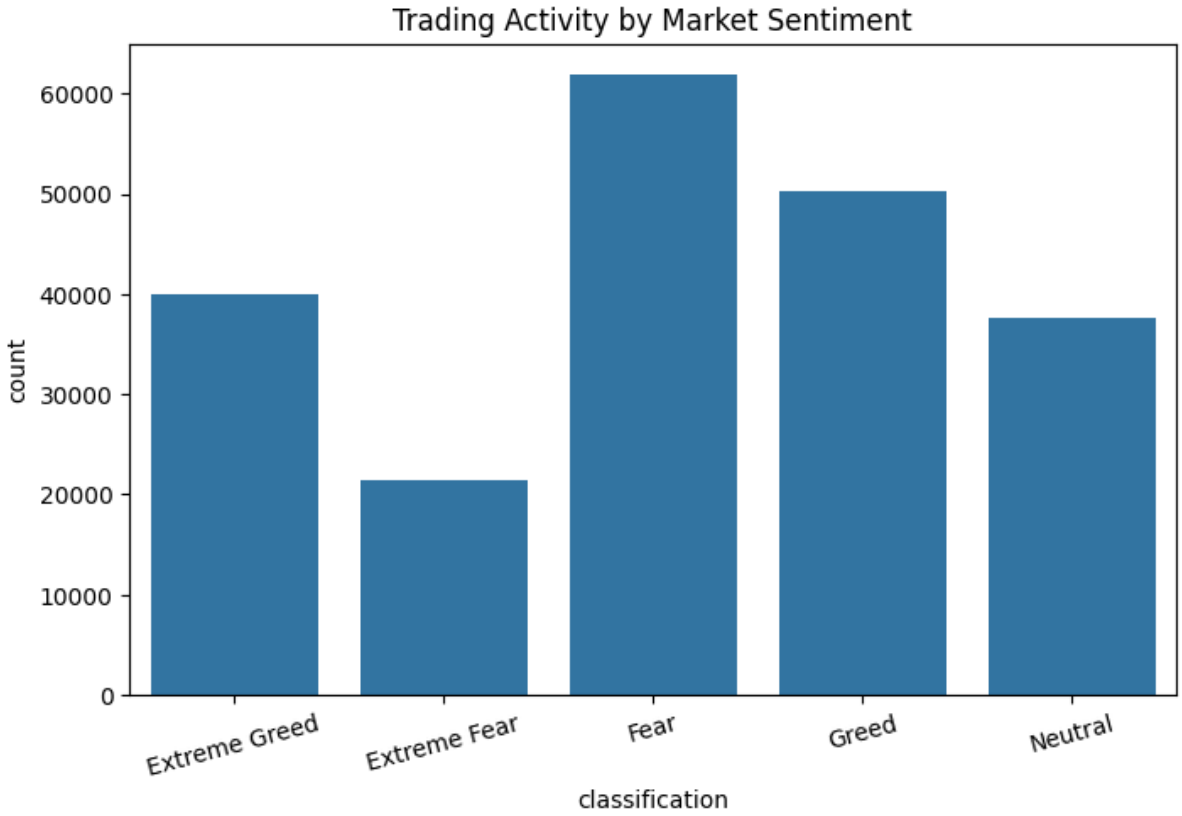
Key Findings

- Fear conditions recorded the highest trading activity with approximately **62,000 trades**.
- Greed conditions also showed elevated participation levels exceeding **50,000 trades**.
- Extreme Fear periods had the lowest trading participation.

Interpretation

High trading activity during Fear and Greed conditions indicates that emotionally driven markets encourage greater participation and speculative behavior. Traders appear more active during volatile market phases where opportunities for rapid price movements are higher.

Lower participation during Extreme Fear periods may reflect market hesitation and reduced confidence.



10. Analysis 4 — Risk Exposure

Risk exposure analysis evaluated how aggressively traders positioned themselves under different market conditions.

Average Risk Exposure by Sentiment

Market Sentiment	Average Risk Exposure
Extreme Greed	3112.25
Neutral	4782.73
Extreme Fear	5349.73
Greed	5736.88

Market Sentiment	Average Risk Exposure
Fear	7816.11

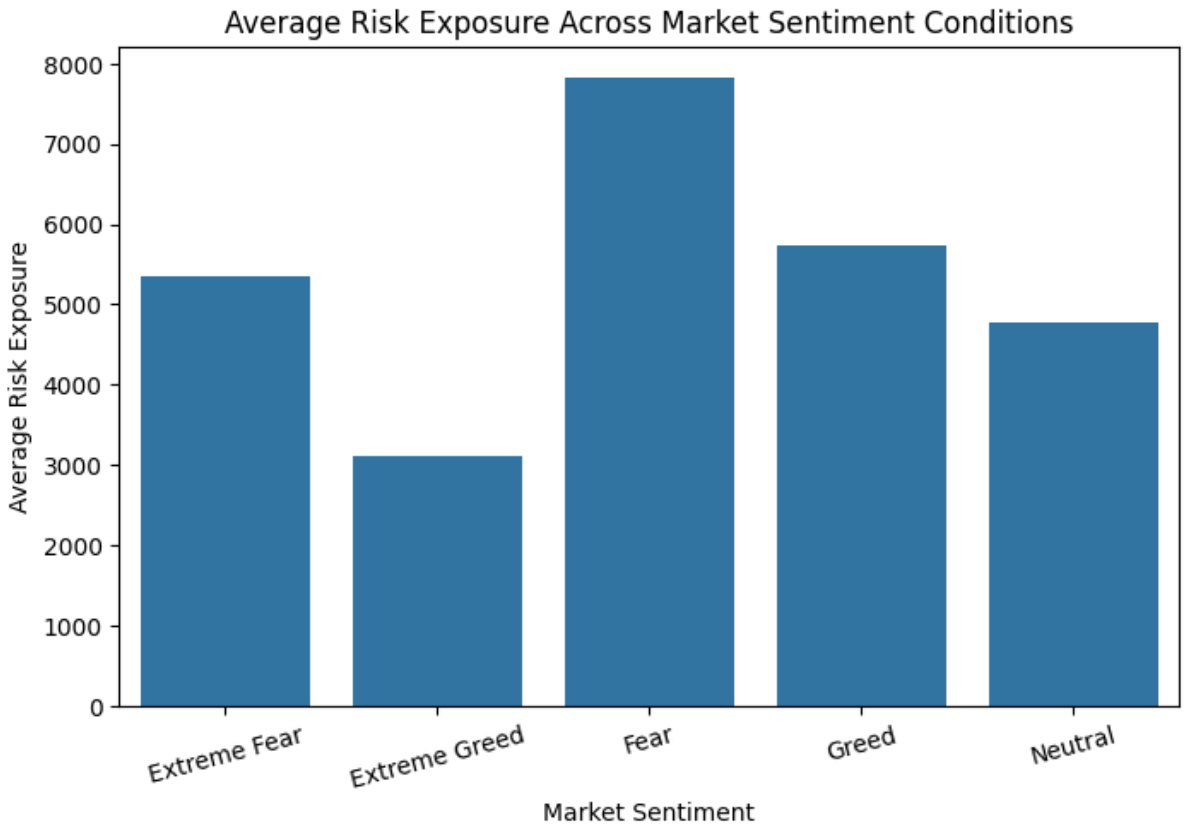
Key Findings

- Fear conditions produced the highest average risk exposure at approximately **7816.11**.
- Greed conditions also showed elevated exposure levels.
- Extreme Greed markets surprisingly showed lower average exposure despite higher profitability.

Interpretation

The results suggest that traders tend to increase position sizes during emotionally volatile periods, particularly during Fear conditions. This behavior may indicate aggressive recovery trading, overconfidence during reversals, or attempts to capitalize on market instability.

Higher exposure significantly increases both profit potential and downside risk.



11. Analysis 5 — Top Trader Analysis

Top-performing traders were analyzed to identify characteristics associated with high profitability.

Top Trader Profitability

Rank	Total Closed PnL
1	2.14 Million
2	1.60 Million
3	0.94 Million
4	0.84 Million
5	0.84 Million

Key Findings

- The highest-performing trader generated cumulative profits exceeding **2.14 million**.
- Multiple traders achieved profits greater than **1 million**.
- Profitability distribution is highly concentrated among a relatively small group of traders.

Interpretation

The findings indicate that successful trading performance is not evenly distributed across participants. A small number of highly skilled or highly capitalized traders capture a disproportionate share of total profits.

This highlights the importance of:

- disciplined execution
- risk management
- strategic position sizing
- market timing

12. Analysis 6 — Correlation Analysis

Correlation analysis was conducted to study relationships between profitability, position size, risk exposure, and trade success.

Correlation Findings

Relationship	Correlation
Closed PnL ↔ Size USD	0.12
Closed PnL ↔ Risk Exposure	0.12
Closed PnL ↔ is_profit	0.094

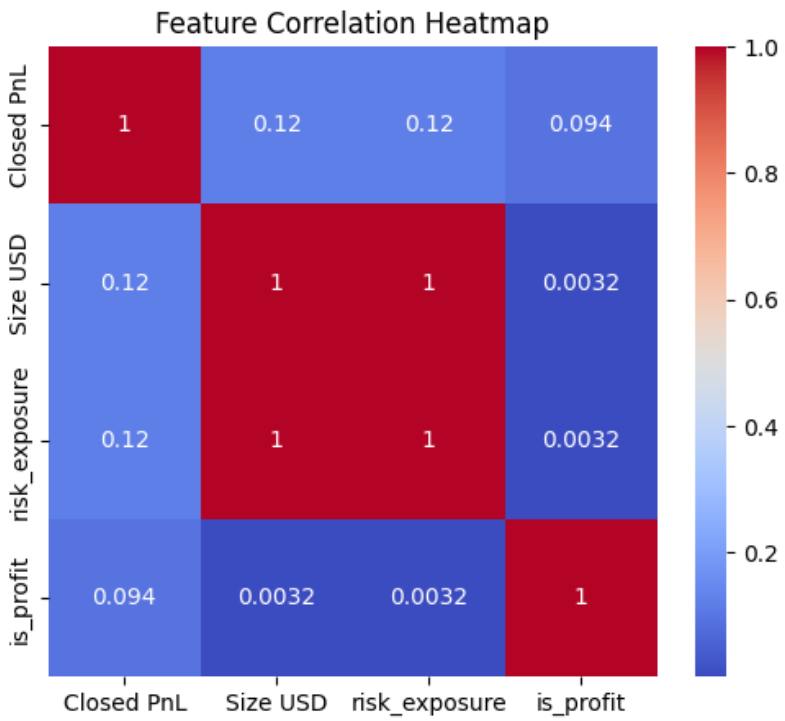
Interpretation

The analysis indicates weak positive relationships between profitability and trading size/risk exposure.

Key observations:

- Larger positions may slightly increase profitability potential
- Excessive exposure does not guarantee trading success
- Position sizing alone is insufficient without effective strategy execution

The results emphasize the importance of balanced risk management rather than simply increasing trade size.



13. Statistical Testing

A statistical t-test was conducted to determine whether profitability differences between Fear and Greed market conditions were statistically significant.

Statistical Result

Metric	Value
P-value	0.0595

Interpretation

The p-value of **0.0595** is slightly above the standard significance threshold of 0.05.

This indicates that:

- profitability differences between Fear and Greed conditions are noticeable,
- but the statistical evidence is not strong enough to conclusively confirm significance at the 95% confidence level.

However, the result remains close to significance and still suggests that market sentiment likely influences trader behavior and trading outcomes.

14. Overall Findings

The analysis produced several important insights:

- Extreme Greed conditions generated the highest trader profitability and win rates.
- Fear conditions showed the highest trading activity and largest risk exposure.
- Market sentiment significantly affects trader behavior and participation.
- Profitability remains highly concentrated among a small group of top traders.
- Larger trade sizes increase volatility but do not guarantee profitability.
- Sentiment-driven markets create both opportunity and elevated risk.

15. Business & Trading Implications

The findings suggest several practical applications for trading strategy development:

- Market sentiment can be used as a supplementary trading signal.
- Risk exposure should be dynamically adjusted according to market conditions.
- Excessive leverage during emotionally volatile periods may increase downside risk.
- Sentiment-aware systems can improve portfolio management and trading discipline.
- Bullish momentum periods may provide stronger short-term trading opportunities.

16. Conclusion

This analysis demonstrates a meaningful relationship between Bitcoin market sentiment and trader behavior.

Bullish sentiment conditions particularly Extreme Greed phases were associated with:

- higher average profitability,
- improved win rates,
- stronger market participation.

Fear conditions generated the highest risk exposure and trading activity, indicating aggressive trader positioning during volatile periods.

Although statistical testing did not produce fully significant results at the 95% confidence level, the findings strongly suggest that market sentiment meaningfully influences trading outcomes, risk-taking behavior, and overall market participation.

The results indicate that sentiment indicators can serve as valuable supplementary tools for:

- trading strategy optimization,
- market timing,
- dynamic risk management,
- and trader behavior analysis in cryptocurrency markets.